@prism project

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The `@prism` project [1] provides 346 color palettes with native support for `CSS`, `Lua`, and LaTeX, to help you create expressive content

Last changes

1.3.0
2026-01-01

Fix.

- `Lua` product: the `use-std.tex` and `use-dark.tex` codes have been fixed.

Break.

- `luadraw` product becomes `Lua` product.
- `Lua` product: the dict-like variable `palNames` has been removed. Therefore, the responsibility of adding this specific variable will lie with `luadraw`.

New.

- New `CSS` and LaTeX products have been coded.
- The `Palettable` palettes have been added.
- Three semantic palettes `TdocBW`, `TdocCol` and `TdocDark` have been created to be used by the `tutodoc` class (LaTeX creation process used).
- Added an `online demo of @prism` (the `CSS` product showcase is used).

Update.

- Manual.
  - Added lists of renamed and removed palettes.
  - Added two appendixes, one with all palettes grouped by user-friendly categories, and another showing similar palettes.
  - Added explanations about the utility of palettes for non-designers.
  - Added explanations about the minimal common optional sets of extra features implemented.
- Better palette references.
- Better showcases.
  - Several category `PDF` files are proposed (one `PDF` per category and theme).
  - 2D chart and Delaunay triangulation added.

[1] The name comes from "***@·esthetic P·roducts*** *for* ***R·epresenting I·nformative S·cientific M·aps***". This name is a double play on words: [1] a prism splits light into an informative spectrum, symbolizing how data are decomposed into meaningful color, and [2] "@" read as "at" indicates where the light meets the prism to be broken down into an informative spectrum.

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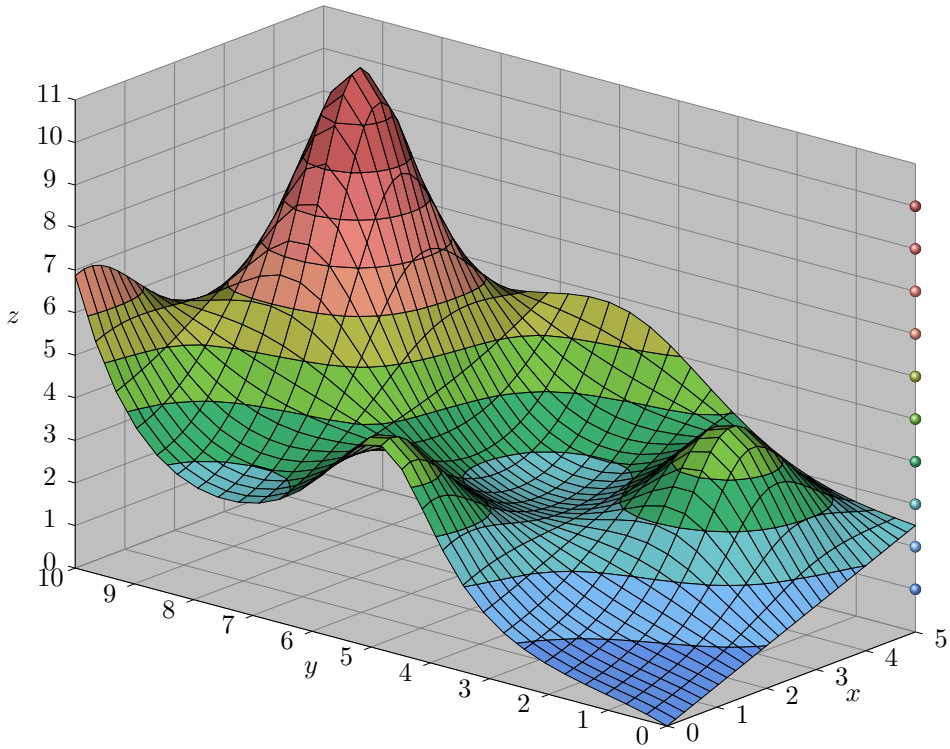
I. Motivations

This project was born out of a desire to enhance `luadraw` with a set of color palettes to easily produce graphics like the 3D plot shown below. Despite the availability of palette generation tools such as `Coolors`, the choice has been made to offer pre-built palettes, because this serves several key purposes for non-designers.

- **Reduced cognitive load:** pre-configuration eliminates design decisions, allowing to focus on content.
- **Consistency:** it becomes easy to ensure visual coherence across documents and projects, regardless of technology.
- **Accessibility compliance:** some pre-configured palettes guarantee colorblind-safe combinations.
- **Perceptual uniformity:** most palettes ensure distinguishable and meaningful colors for data visualization, while others serve artistic purposes.

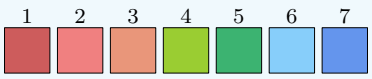


Technically, to generate the graphics in this section, `luadraw` is provided with a finite list of colors and uses linear interpolations to calculate the intermediate color values. In the example above, the finite color palette used is defined as follows.

Using this palette, `luadraw` can produce the following spectrum, allowing us to create the graphic above.

Note.

Using the `Lua` *implementation of* `@prism`*, see the section V-4, via* `luadraw`*, we can create the palettes below made from the previous one named* `'GeoRainbow'`*. Each instruction used is given below each palette.*

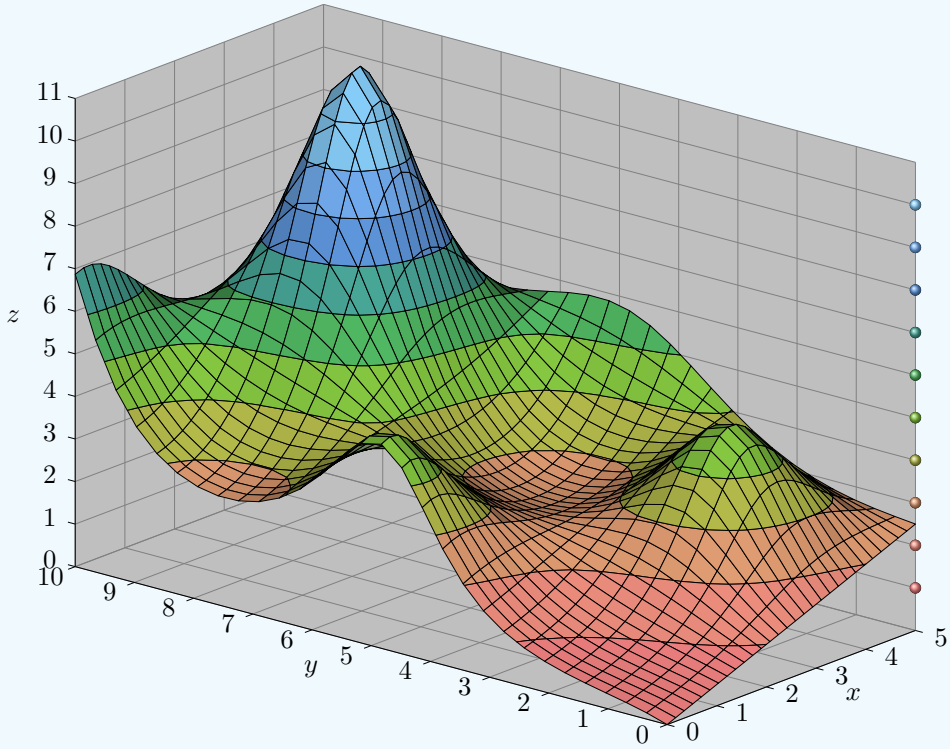



```
getPal('GeoRainbow', {reverse = true})
```

```
getPal('GeoRainbow', {shift = 3})
```

`getPal('GeoRainbow', {extract = {7, 1, 4}})`

This features provide remarkable creative flexibility: with the same surface as before, using the setting `getPal('GeoRainbow', {extract = {6, 5, 4, 3, 1, 2}})` *instead of* `getPal('GeoRainbow')`*, we change the visual tone, shifting from a seaside feel to a snow-covered world.*



II. Where do the color palettes come from?

`@prism` includes some original creations, but most color palettes are derived from the projects listed below.

- `Asymptote` is used, but currently offers nothing beyond `Matplotlib` (despite different implementations).
- `CarbonPlan` provides palettes designed for data visualization..
- `CARTOColors` palettes are extracted from `Palettable` project.
- `CMasher` provides palettes designed for data visualization..
- `cmocean` palettes are extracted from `Palettable` project.
- `Colorbrewer` provides professional color palettes for mapping and data visualization.
- `Light and Bartlein` palettes are extracted from `Palettable` project.
- `Matplotlib` compiles color maps from diverse projects.
- `MyCarta` palettes are extracted from `Palettable` project.
- `Plotly` palettes are extracted from `Palettable` project.
- `Scientific Coulour Maps` provides palettes designed for colorblind accessibility.
- `Tableau` palettes are extracted from `Palettable` project.
- `Wes Anderson Palettes` palettes are extracted from `Palettable` project.

We retain only palettes that comply with the following rules.

- **No repetition.** Unlike `Matplotlib`,[2] `@prism` use a one-to-one map from names to palettes.[3]

[2] Some `Matplotlib` palettes are duplicated, likely for historical reasons.
[3] This does not preclude the existence of palettes with different names that are visually similar, or even looks identical. This

- **No reversed versions.** Unlike `Matplotlib`,[4] `@prism` never includes reversed palettes as fixed data.
- **No shifted versions.** Unlike `Colormaps`,[5] `@prism` never includes shifted palettes as fixed data (most implementations offer an API that enables this and other transformations through configurable parameters).
- **Reasonable palette sizes.** Palettes with more than 512 colors are ignored (e.g., `CMasher` proposes 'neutral' palette, which consists of 2560 colors). This avoids potential issues with certain technologies, such as LaTeX, and frankly, offers a level of precision that is far beyond the needs of us mere mortals — or at least, those of us who aren't astronomers.

Note.

Adding new palettes to `@prism` is straightforward (no coding skills required). See section VI-2 to get started.

III. Reuse from...

Here are the key points to remember when using palettes similar to those offered by projects listed in the section II.

1. `Matplotlib` palettes with a name ending with `_r` (reversed color order), and shifted palettes[6] are not included in `@prism`. In rare cases, some palettes must be excluded because they seem superfluous or are too large. All the excluded palettes are listed in Appendix 3 (page 18).
2. `@prism` uses 'semantic' `CamelCase` naming. Therefore, palette names such as `berlin`, `gist_heat` and `Pinkgrey` become `Berlin`, `GistHeat` and `PinkGrey`. To avoid naming collisions, certain palettes have been renamed : a full list of these changes is provided in Appendix 2 (page 16). You can see the full list of names in Appendix 1 (page 13).

Caution.

Most `@prism` implementations add the `pal` prefix to standardized `CamelCase` names. See the section V.

Note.

Most `@prism` implementations provide methods to easily obtain reversed palettes, sub-palettes, and color-shifted palettes. See the section V.

IV. How to choose a palette?

Here are a few tips to help you select your color palette.

1. An `online demo of @prism`[7] proposes an intuitive way to look for a palettes with a random button to discover unknown palettes.
2. Appendix 4 (page 19) presents all palettes, organized by category, along with a visualization of their continuous spectra (size > 40), or their corresponding discrete palette definitions (size ≤ 40).

V. Supported implementations

All implementations are located in the `products` folder. Each implementation provides the following features.

- Modular palette formats (one file per palette).
- Palette definitions in both high-fidelity (original size) and small (currently 40 colors) size.

is often justified by the number of colors used; a higher color depth helps prevent artifacts when generating high-resolution graphics.

[4] Most `Matplotlib` color maps have a reversed version named with the `_r` suffix, possibly for performance reasons.

[5] These palettes use an integer suffix directly following the original palette name.

[6] By design, `Colormaps` proposes some shifted palettes.

[7] `https://projetmbc.github.io/for-writing/`

Most implementations also feature the API explained below.

- Select specific colors from an existing palette using their indices.
- Shift the palette left (negative value) or right (positive value) by any number of steps.
- Reverse the order of the colors.

Important.

To explain how new palettes can be built, we will refer to the colors in the standard palette as `coul_1`, `coul_2`, etc., and suppose that the extracted indices are `{1, 3, 6, 9}`, the shift used is 1+, and the `reverse` option is enabled. The new palette will then be built sequentially as follows: first `{coul_1, coul_3, coul_6, coul_9}` (extraction), second `{coul_9, coul_1, coul_3, coul_6}` (shift to the right), and finally `{coul_6, coul_3, coul_1, coul_9}` (reverse).

Note.

Extra features are limited to discrete palette operations. For example, color interpolation is not provided, as this is usually handled out of the box by visualization and formatting tools.

1. JSON, the versatile default format

The `JSON` product allows unsupported programming languages to integrate `@prism` palettes easily. Palettes are defined as shown in the following example.

```
[
  [0.4980392157, 0.7882352941, 0.4980392157],
  [0.7450980392, 0.6823529412, 0.831372549],
  [0.9921568627, 0.7529411765, 0.5254901961],
  ...
]
```

2. CSS

Note.

The `HTML`, `CSS`, and `JavaScript` files for product development and demonstration were created using `Claude` and `Gemini` AI assistants.

Each palette color is defined as an individual variable named according to the pattern `--pal<name>-<nb>`, where `<name>` is the standard palette name and `<nb>` is the desired index ranging. Each palette color variable is defined as an `RGB` value using percentage notation. For example, the file `palettes-hf/GistHeat.css` looks like the following partial code.

```
:root {
  --palGistHeat-1: rgb(0% 0% 0%);
  --palGistHeat-2: rgb(0.59% 0% 0%);
  /* Other RBG colors.*/
  --palGistHeat-256: rgb(100% 100% 100%);
}
```

Here is one first possible use case.

```
.warning-text {
  color: var(--palGistHeat-3);
}

.gist-heat-gradient {
  background: linear-gradient(
    90deg,
    var(--palGistHeat-1),
    var(--palGistHeat-64),
    var(--palGistHeat-128),
    var(--palGistHeat-256)
```

```
  );
}
```

The example below demonstrates creating gradient variables through selective color extraction and custom reordering.

```
:root {
  --transformed-gist-heat-gradient: linear-gradient(
    90deg,
    var(--palGistHeat-6),
    var(--palGistHeat-3),
    var(--palGistHeat-9),
    var(--palGistHeat-1)
  );
}

.transformed-gist-heat-gradient {
  background: var(--transformed-gist-heat-gradient);
}
```

3. LaTeX

a. Basic use

Accessing a single palette color is straightforward: use `\palUse{<name>}{<index>}` where `<name>` is the standard palette name (without prefix), and `<index>` is the color number (ranging from 1 to 10). For example, `\palUse{GistHeat}{8}` is the eighth color of the `GistHeat` palette, an `xcolor` format color that can be easily used as shown in the following compilable example.

```
\documentclass{article}

% Load the palette (LaTeX can't work with all the palettes).
\usepackage{palettes-hf/GistHeat}

% Load the palette API.
\usepackage{palapi}

\usepackage{tikz}

\begin{document}

\textcolor{\palUse{GistHeat}{8}}{\bfseries Colored text.}

Representation of the first ten palette colors.

\begin{tikzpicture}
  \foreach \i in {1,...,10} {
    \fill[\palUse{GistHeat}{\i}]
      (1.25*\i - 1, 0) rectangle (1.25*\i, 1);
  }
\end{tikzpicture}

\end{document}
```

b. Creating palettes from scratch

For creating new palettes manually, the following high-level commands are available.

1. `\palCreateFromNames` works with a comma separated list of named colors, while `\palCreateFromRGB` creates a palette by entering it as an array-like variable.
2. `\palSize{<name>}` returns the palette size (useful for loops, for example).

The following example demonstrates the `\palCreateFromRGB` and `\palCreateFromNames` commands.

```
\usepackage{palapi}

\usepackage[svgnames]{xcolor}

\palCreateFromRGB{MyRGBPal}{
  {0.0, 0.0, 0.0},
  {0.4, 0.0, 0.2},
  {0.8, 0.2, 0.0},
}

\palCreateFromNames{MyNameUsePal}{
  YellowGreen,
  green!60!black,
  LimeGreen!80,
}
```

Note.

All built-in palettes are created using the `\palCreateFromRGB` macro.

A lower-level approach is also available through the following commands.

1. `\palNew{<name>}` initializes a new (empty) palette.
2. `\palAddNames{<name>}{<color-using-names>}` appends a color defined with named colors to the palette.
3. `\palAddRGB{<name>}{<r>, <g>, <b>}` appends an `RGB` color to the palette, where `<r>`, `<g>`, and `<b>` are decimal values ranging from 0 to 1.

The following example demonstrates the flexibility offered by these low-level commands.

```
\usepackage{palapi}

\usepackage[svgnames]{xcolor}

\palNew{LowLevelPal}
\palAddNames{LowLevelPal}{IndianRed}
\palAddRGB{LowLevelPal}{0.0, 0.0, 0.0}
\palAddNames{LowLevelPal}{green!60!black}
\palAddRGB{LowLevelPal}{0.8, 0.2, 0.0}
```

c. Creating palettes from existing ones

Building new palettes by transforming existing ones can be achieved using the `\palCreateFromPal` command, which has the signature `\palCreateFromPal{<new-name>}[<options>]{<existing-name>}`. The following example shows how to do this (all options are used).

```
\usepackage{palettes-hf/Blackbody}
\usepackage{palapi}

\palCreateFromPal{BlackbodyTransformed}[
  extract = {1, 3, 6, 9},
  shift   = 1,
  reverse
]{Blackbody}
```

Tip.

`\palCreateFromPal{<new-name>}{<existing-name>}` builds a copy of an existing palette.

d. Retrieving the internal definition of a palette

The internally stored definition of a palette named `MyPal`, for example, is `\g_palette_MyPal_seq` which is a LaTeX3 variable (keep in mind the pattern `\g_palette_PaletteName_seq`).

Note.

Variables of type `\g_palette_PaletteName_seq` *are not used internally to retrieve the colors themselves; they are only there for technical reasons related to the development process of new palettes via LaTeX.*

4. Lua

Note.

Initially, the `@prism` *project was created to provide ready-to-use color palettes for* `luadraw`*, a package that greatly facilitates the creation of high-quality 2D and 3D plots using LuaLaTeX and* `TikZ`*.*

a. Basic use

The `Lua` palette variables are named using the prefix `pal`. They are arrays of arrays of three floats (making it straightforward to use a color from a palette). For example, the definition of `palGistHeat` looks like the following partial code.

```
palGistHeat = {
    {0.0, 0.0, 0.0},
    {0.105882, 0.0, 0.0},
    {0.211764, 0.0, 0.0},
    -- Other RBG colors.
}
```

b. Creating palettes from existing ones

The `palCreateFromPal` function provides options to build new palettes by transforming existing ones. The following example shows how to do this (all options are used).

```
palBlackbodyTransformed = palCreateFromPal(
    palBlackbody,
    {
        extract = {2, 5, 8, 9},
        shift   = 1,
        reverse = true
    }
)
```

VI. Contribute via Git

Caution.

Never use the `main` *branch, which is for freezing the latest stable versions of each projects in the mono repository* `https://github.com/projetmbc/for-writing`*.*

1. Complete the translations

Important.

Although we're going to explain how to translate the documentation, it doesn't seem relevant to do so, as English should suffice these days.

The translations are roughly organized as in figure 1 where just the important folders for the translations have been "*opened*".[8] **A little further down, the section VI-1-e explains how to add new translations**.

a. The en folder

This folder, managed by the author of `@prism`, contains files easy to translate even if you're not a coder.

[8] This was the organization on October 26, 2025.

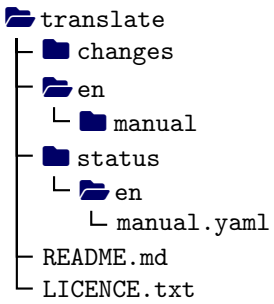


Figure 1: Simplified view of the translation folder

b. The changes folder

This folder is a communication tool where important changes are indicated without dwelling on minor modifications specific to one or more translations.

c. The status folder

This folder is used to keep track of translations from the project's point of view. Everything is done via well-commented `YAML` files, readable by a non-coder.

d. The README.md and LICENCE.txt files

The `LICENCE.txt` file is aptly named, while the `README.md` file takes up in English the important points of what is said in this section about new translations.

e. New translations

Note.

The folder `manual` *is reserved for documentation. It contains* `TEX` *files that can be compiled directly for real-time validation of translations.*

Warning.

Only start from the `en` *folder, as it's the responsibility of the* `@prism` *author.*

Let's say you want to add support for Italian.[9] To do this, you must use `Git` as follows.

1. Via `https://github.com/projetmbc/for-writing/tree/aprism/@prism`, recover the entire project folder. Do not use the `main` branch, which is used to freeze the latest stable versions of all the projects in the mono repository `https://github.com/projetmbc/for-writing`.
2. In the `@prism/contrib/translate` folder, create an `it` copy of the `en` folder, where `it` is the short name of the language documented in the page "*IIETF language tag*" from `Wikipedia`.
3. Once the translation is complete in the `it` folder, share it via `https://github.com/projetmbc/for-writing/tree/aprism/@prism` using a classic `git push`.

2. Improving the source code

Participation as a coder is made via the repository `https://github.com/projetmbc/for-writing/tree/aprism/@prism` corresponding to the `@prism` development branch. Here is what you can do, details can be found in the file `https://github.com/projetmbc/for-writing/blob/aprism/@prism/contrib/products/README.md`.

1. Create new palettes within an existing implementation. No coding skills required.
2. Propose a new implementation in your favorite programming language.
3. Combine both approaches.

[9] As mentioned above, there is no real need for the `doc` folder.

VII. History

1.3.0
2026-01-01

Fix.

- Lua product: the `use-std.tex` and `use-dark.tex` codes have been fixed.

Break.

- luadraw product becomes Lua product.
- Lua product: the dict-like variable palNames has been removed. Therefore, the responsibility of adding this specific variable will lie with luadraw.

New.

- New CSS and LaTeX products have been coded.
- The Palettable palettes have been added.
- Three semantic palettes `TdocBW`, `TdocCol` and `TdocDark` have been created to be used by the tutodoc class (LaTeX creation process used).
- Added an online demo of @prism (the CSS product showcase is used).

Update.

- Manual.
  - Added lists of renamed and removed palettes.
  - Added two appendixes, one with all palettes grouped by user-friendly categories, and another showing similar palettes.
  - Added explanations about the utility of palettes for non-designers.
  - Added explanations about the minimal common optional sets of extra features implemented.
- Better palette references.
- Better showcases.
  - Several category PDF files are proposed (one PDF per category and theme).
  - 2D chart and Delaunay triangulation added.

1.2.1
2025-11-09

Fix.

- Equal palettes: the floating point equality uses now a correct tolerance.

Break.

- Palettes: the extra `Greys` has been removed (it is equal to `Grays`).

New.

- Similar palettes: two PDF files show similar palettes in standard and black modes (semi-automated process used).

Update.

- luadraw product: the associative array palNames has been added for compatibility reasons with the luadraw package.
- `BlindFish` palette: the last color variation has been made smoother (luadraw process used).

1.2.0
2025-10-29

Break.

- Palettes: all final palettes now consist of 10 colors.
- luadraw products: the getPal dictionary array has been converted into a function accepting string palette names (with or without pal prefix). See below.

New.

- Palettes.
  - Added `Lemon` and `ShiftRainbow` palettes (luadraw creation process used).
  - Added 37 palettes from the Scientific Coulour Maps project.
- luadraw product: accessing a palette and creating new ones can be made using the getPal function which has an optional argument options (dict-like array) with the following keys and their values.
  - extract: a list of non-zero integers used to extract specific colors from the palette (the order is preserved).
  - reverse: a boolean value indicating whether to reverse the palette color order (false by default).
  - shift: an integer value for applying a circular color shift to the palette.
- Documentations
  - Added English PDF manual.
  - Showcase: two PDF files demonstrate the use of each palette (white and dark modes).

1.1.0
2025-10-14

Break.

- Duplicate palettes and those that are reverse of others are ignored (strict equalities only).

New.

- New palettes added: `BurningGrass`, `GeoRainbow` and `PastelRainbow` (`luadraw` creation process used).
- The `luadraw` palette product has a new dictionary like variable `getPal` to access a palette using its name (as a string variable).

Update.

- Palette contributions: in the mandatory `extend.py` file, the `build_code` function must work with the dictionary of all the palettes, and manage a credit to the `@prism` project.

1.0.0
2025-10-11

First public version of the project.

Appendix 1 – All palette names

Important.

The palette names in this appendix are standard, but some `@prism` implementations add a specific prefix, and the very last new palettes are marked by ★ .

A

Accent
Acton
AfmHot
AgGrnYl
AgSunset
Algae
Alphabet
Amber ★
Amethyst ★
Amp
Antique
Apple ★
Aquatic1
Aquatic2
Aquatic3
Arctic ★
ArmyRose
Autumn

B

Balance ★
Bam
Bamako
BamO
Batlow
BatlowK
BatlowW
Berlin
Bilbao
Binary
Blackbody
BlindFish
BlueDarkOrange ★
BlueDarkRed12
BlueDarkRed18
BlueGray
BlueGreen
BlueGrey ★
BlueOrange ★
BlueOrangeRed
BlueRed
BlueRedPLY
Blues
Blues7
Blues10
BluesCP ★
BluGrn
BluYl
Bold
Bone
BrBG
BRG
Broc
BrocO
BrownBlue10
BrownBlue12
BrwnYl
Bubblegum ★
Buda
BuGn
Bukavu
BuPu
Burg
BurgYl
BurningGrass
BWR

C

Cavalcanti
Chevalier
Chroma ★
Cividis
Classic ★
CMRmap
ColorBlind
Cool
CoolCP ★
CoolWarm
Copper
CopperCM ★
Cork
CorkO
Cosmic ★
Cube1
Cubehelix
Cubehelix1
Cubehelix2
Cubehelix3
CubeYF
Curl

D

Darjeeling1
Darjeeling2
Darjeeling3
Darjeeling4
Dark2
Dark24
DarkMint
Davos
Deep
Delta
Dense
Devon
Dusk ★

E

Earth
EarthCP ★
Eclipse ★
Electric
Ember ★
Emerald ★
Emergency ★
Emrld

F

Fall
FallCM ★
FantasticFox1
FantasticFox2
Fes
Fire ★
Flag
Flamingo ★
Freeze ★
Fusion ★

G

G10
GasFlame
Gem ★
GeoRainbow
Geyser
GhostLight ★
GistEarth

GistGray ★
GistHeat
GistNcar
GistRainbow
GistStern
Glasgow
GnBu
Gnuplot
Gnuplot2
Gothic ★
GrandBudapest1
GrandBudapest2
GrandBudapest3
GrandBudapest4
GrandBudapest5
Gray ★
GrayC
GreenGrey ★
GreenMagenta
GreenOrange
Greens
GreensCP ★
Greys ★
GreysCP ★
Guppy ★

H

Haline
Hawaii
Heart ★
Holly ★
Horizon ★
Hot
HotPLY ★
HSV

I

Ice
IceBurn ★
Imola
Inferno
Infinity ★
IsleOfDogs1
IsleOfDogs2
IsleOfDogs3

J

Jet
JetPLY ★
JimSpecial
Jungle ★

L

Lajolla
Lapaz
Lavender ★
Lemon
Light24
Lilac ★
LinearL
Lipari
Lisbon

M

Magenta
Magma
Managua
Margot1
Margot2
Margot3
Matter
Mendl
MidGray ★
Mint
Moonrise1
Moonrise2
Moonrise3
Moonrise4
Moonrise5
Moonrise6
Moonrise7

N

Navia
NaviaW
Neon ★
NipySpectral
Nuclear ★
Nuuk

O

Ocean
OceanCM ★
OkabeIto ★
Oleron
OrangeBlue ★
OrangeGrey ★
Oranges
OrangesCP ★
OrRd
OrYel
Oslo
Oxy

P

Paired
Pastel
Pastel1
Pastel2
PastelRainbow
Peach
Pepper ★
PerceptualRainbow
Petroff6 ★
Petroff8 ★
Petroff10 ★
Phase
Picnic
Pink
PinkGreen ★
PinkGrey ★
Pinks ★
PinkYl
PiYG
Plasma
Plotly
Portland
PrGn
Pride ★
Prinsenvlag ★
Prism
PrismCC
PuBu
PuBuGn
PuOr
PuRd
Purp
Purple
PurpleGray
PurpleGrey ★
Purples
PurplesCP ★
PurpOr

R

Rainbow
RainbowCP ★
RainbowPLY
Rainforest ★

RdBu
RdGy
RdPu
RdYlBu
RdYlGn
Red
RedGrey ★
RedOr
Reds
RedsCP ★
RedShift ★
RedTeal ★
RedYellowBlue
Roma
RomaO
Royal1
Royal2
Royal3

S

Safe
Sapphire ★
Seasons ★
Seaweed ★
Seismic
Sepia ★
Set1
Set2
Set3
ShiftRainbow
SineBow ★
Solar
Spectral
Speed
Spring
Summer
SunBurst ★
Sunset
SunsetDark
Swamp ★

T

T10
Tab20
Tab20b
Tab20c
TableauLight
TableauMedium
TdocBW
TdocCol
TdocDark
Teal
TealGrey ★
TealGrn
TealRose
Teals ★
Tempo
Temps
Terrain
Thermal
Tofino
Tokyo
Torch ★
Toxic ★
TrafficLight
Tree ★
Tropic
Tropical ★
Turbid
Turbo
Turku
Twilight

V

Vanimo
Vik
VikO
Viola ★
Viridis
Vivid
Voltage ★

W

Warm ★
Water ★
Waterlily ★
Watermelon ★
Wildfire ★
Wind ★
Winter
Wistia

Y

YellowGrey ★
YellowPurple ★
Yellows ★
YlGn
YlGnBu
YlOrBr
YlOrRd

Z

Zissou

Appendix 2 – Renamed palettes

Important.

The palette names in this appendix are standard, but some `@prism` implementations add a specific prefix.

The following palettes have been renamed. We use $\boxed{\Rightarrow}$ to denote these changes, with the new `@prism` names on the right.

Table 1: Renamed palettes

CarbonPlan	`Bluegrey`	⇒	`BlueGrey`
	`Blues`	⇒	`BluesCP`
	`Cool`	⇒	`CoolCP`
	`Earth`	⇒	`EarthCP`
	`Greengrey`	⇒	`GreenGrey`
	`Greens`	⇒	`GreensCP`
	`Greys`	⇒	`GreysCP`
	`Orangeblue`	⇒	`OrangeBlue`
	`Orangegrey`	⇒	`OrangeGrey`
	`Oranges`	⇒	`OrangesCP`
	`Pinkgreen`	⇒	`PinkGreen`
	`Pinkgrey`	⇒	`PinkGrey`
	`Purplegrey`	⇒	`PurpleGrey`
	`Purples`	⇒	`PurplesCP`
	`Rainbow`	⇒	`RainbowCP`
	`Redgrey`	⇒	`RedGrey`
	`Reds`	⇒	`RedsCP`
	`Redteal`	⇒	`RedTeal`
	`Sinebow`	⇒	`SineBow`
	`Tealgrey`	⇒	`TealGrey`
	`Yellowgrey`	⇒	`YellowGrey`
	`Yellowpurple`	⇒	`YellowPurple`
CARTOColors	`Prism`	⇒	`PrismCC`
CMasher	`Copper`	⇒	`CopperCM`
	`Fall`	⇒	`FallCM`
	`Ghostlight`	⇒	`GhostLight`
	`Iceburn`	⇒	`IceBurn`
	`Ocean`	⇒	`OceanCM`
	`Redshift`	⇒	`RedShift`
	`Sunburst`	⇒	`SunBurst`
Colorbrewer	`PRGn`	⇒	`PrGn`
	`YlOrRd`	⇒	`YlOrRdCBW`
Light and Bartlein	`BlueDarkOrange18`	⇒	`BlueDarkOrange`
	`BlueOrange12`	⇒	`BlueOrange`
Matplotlib	`Afmhot`	⇒	`AfmHot`

Continued on next page

Table 1: Renamed palettes (Continued)

	`Brg`	⇒	`BRG`
	`Bwr`	⇒	`BWR`
	`Coolwarm`	⇒	`CoolWarm`
	`Hsv`	⇒	`HSV`
Plotly	`Bluered`	⇒	`BlueRedPLY`
	`Hot`	⇒	`HotPLY`
	`Jet`	⇒	`JetPLY`
	`Rainbow`	⇒	`RainbowPLY`
Tableau	`Gray`	⇒	`MidGray`

Appendix 3 – Ignored palettes

Important.

The palette names in this appendix are standard, but some `@prism` implementations add a specific prefix.

Rebuildable palettes

The following palettes are excluded as sublists[10] or shifts of existing ones, or due to direct or reversed duplication. Note that exact duplicates – sharing both name and colors – are omitted entirely. We identify relation between palettes using the following notations.

- Sublist: $\boxed{\prec}$
- Shifted: $\boxed{\ll}$
- Equality: $\boxed{=}$
- Reversal: $\boxed{\rightleftharpoons}$

In these cases, the rightmost palette is the one retained in `@prism`. For example, this table indicates that `YlOrRd` from `Colorbrewer` uses a sublist of colors from the `@prism` palette `YlOrRd`.

Table 2: Rebuildable palettes

Colorbrewer	`YlOrRd`	$\prec$	`YlOrRd`
Light and Bartlein	`BlueDarkOrange12`	$\prec$	`BlueDarkOrange`
	`BlueOrange10`	$\prec$	`BlueOrange`
	`BlueOrange8`	$\prec$	`BlueOrange`
Matplotlib	`Tab10`	$\prec$	`Tab20`
	`TwilightShifted`	$\ll$	`Twilight`
	`Gray`	$\rightleftharpoons$	`Binary`
	`GistYarg`	$\rightleftharpoons$	`GistGray`
Plotly	`D3`	$\prec$	`Tab20`
Tableau	`Tableau`	$=$	`Tab20`

Excluded by design

The following palettes have been intentionally excluded. Please refer to the explanation below.

Table 3: Excluded palettes

CMasher	`Neutral`	Too big (size > 512).
	`Savanna`	Looks completely black.

[10] A palette is defined as a sublist if it is formed by removing colors while preserving the relative order of the remaining ones.

Appendix 4 – Palettes by category

Important.

The palette names in this appendix are standard, but some `@prism` implementations add a specific prefix.

Note (Qualitative and semantic palettes).

The threshold of 40 colors has been established for qualitative and semantic palettes. The following lines outline the rationale for this limit.

- *In qualitative color palettes, each color represents a single data point or a group of data. Experience shows that about ten colors are more than enough.*
- *Semantic palettes associate a color with a specific context, such as warning levels (see the three frames in this document). Choosing 40 colors allows for 10 different contexts, each with 4 sub-categories. Here are some examples of sub-categories.*

  10-19 ***Initial / Idle:*** *default states such as "To Do", "Clickable", or "To be analyzed".*
  20-29 ***Active / Transition:*** *states in flux, including "In Progress" or UI "Hover" effects.*
  30-39 ***Success / Final:*** *terminal positive states like "Done", "Clicked", or "Validated".*
  40-49 ***Alert / Rejection:*** *error handling and negative outcomes, e.g., "Error", "Inactive", or "Rejected".*

Important.

Categories were generated semi-automatically, followed by manual selection to obtain relevant choices.

Colorblind palettes

Colorblind palettes provide high-contrast hues specifically optimized for users with color vision deficiency.

Acton

Bam

Bamako

BamO

Batlow

BatlowK

BatlowW

Berlin

Bilbao

Broc

BrocO

Buda

Bukavu

Cork

CorkO

Cubehelix

Davos

Devon

Fes

Glasgow

GrayC

Hawaii

Imola

Lajolla

Lapaz

Lipari

Lisbon

Managua

Navia

NaviaW

Nuuk

OkabeIto

Oleron

Oslo

Petroff6

Petroff8

Petroff10

Roma

RomaO

Safe

Tofino

Tokyo

Turku

Vanimo

Vik

VikO

Wistia

Cyclic palettes

Cyclic palettes create seamless loops for periodic data, such as angles, seasons, or 24-hour cycles.

CopperCM

Emergency

Flag

Infinity

RainbowCP

Seasons

SineBow

Dark palettes

Dark palettes offer high-contrast schemes specifically optimized for terminal interfaces and dark-mode backgrounds.

Autumn

BWR

Cool

GistRainbow

HSV

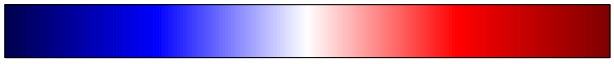
Jet

Lemon

Prism

Rainbow

Seismic



Spring

Summer

Tab20

Tab20b

Tab20c

TdocDark

Terrain

Viridis

Winter

YlOrRd

Divergent palettes

Divergent palettes highlight deviations from one or more midpoints toward contrasting extremes.

ArmyRose

Balance

BlindFish

BlueDarkOrange

BlueDarkRed12

BlueDarkRed18

BlueGray

BlueGreen

BlueGrey

BlueOrange

BlueOrangeRed

BrBG

BrownBlue10

BrownBlue12

BurningGrass

BWR

CoolWarm

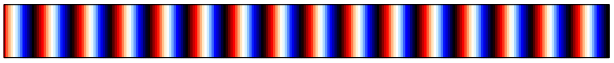
Curl

Delta

Earth

Fall

Flag



Fusion

Geyser

GistNcar

GistRainbow

GistStern

GreenGrey

GreenMagenta

Guppy

Holly

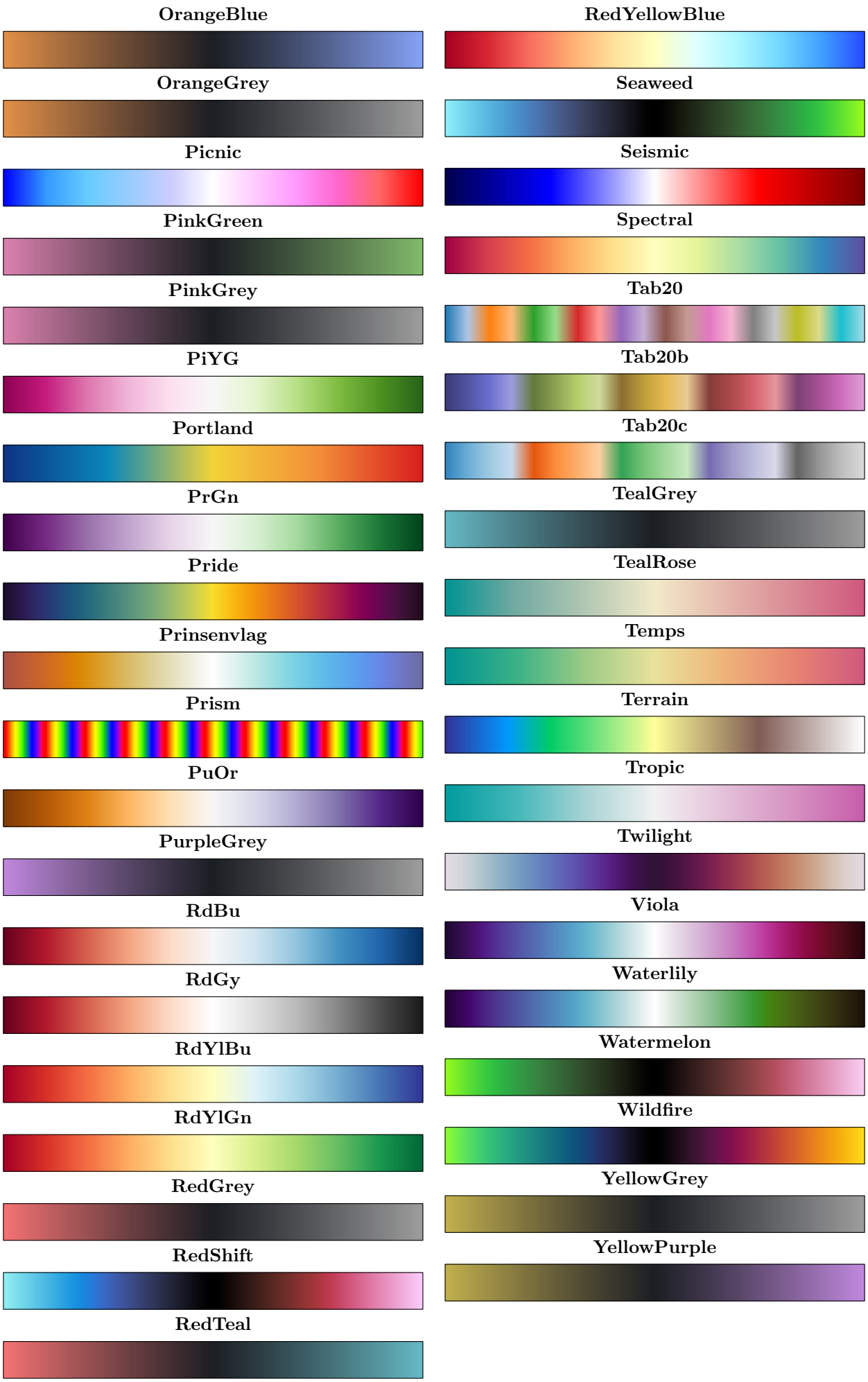
HSV

IceBurn

Jet

NipySpectral

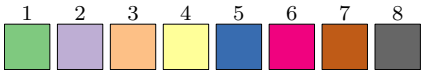
Ocean


OrangeBlue
OrangeGrey
Picnic
PinkGreen
PinkGrey
PiYG
Portland
PrGn
Pride
Prinsenvlag
Prism
PuOr
PurpleGrey
RdBu
RdGy
RdYlBu
RdYlGn
RedGrey
RedShift
RedTeal
RedYellowBlue
Seaweed
Seismic
Spectral
Tab20
Tab20b
Tab20c
TealGrey
TealRose
Temps
Terrain
Tropic
Twilight
Viola
Waterlily
Watermelon
Wildfire
YellowGrey
YellowPurple

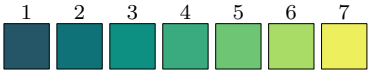
Qualitative palettes

Qualitative palettes use distinct hues to separate categorical data without implying any order or hierarchy.

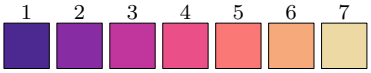
Accent



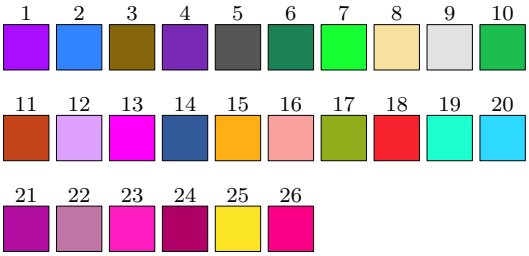
AgGrnYl



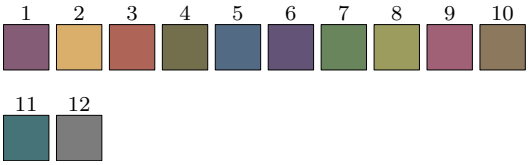
AgSunset



Alphabet

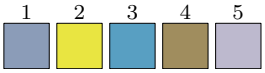


Antique

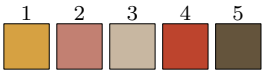


Aquatic1

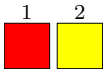
Aquatic2



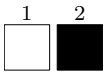
Aquatic3



Autumn

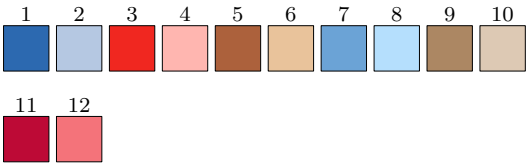


Binary

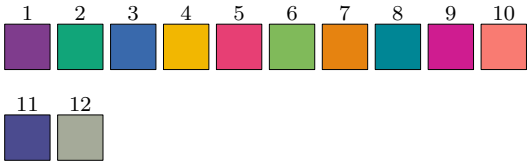


BlindFish

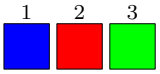
BlueRed



Bold

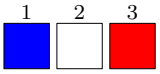


BRG

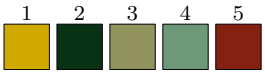


BurningGrass

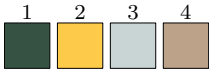
BWR



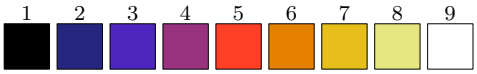
Cavalcanti



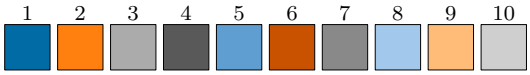
Chevalier



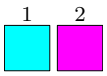
CMRmap



ColorBlind

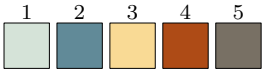


Cool

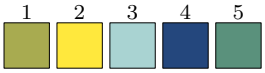


Darjeeling1

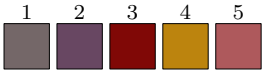
Darjeeling2



Darjeeling3

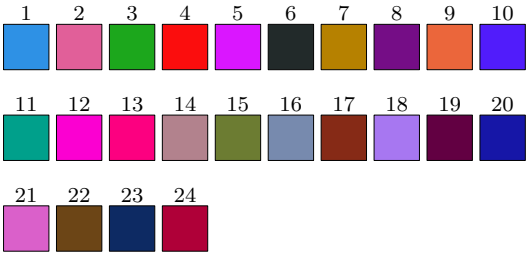


Darjeeling4

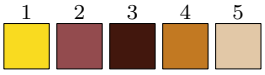


Dark2

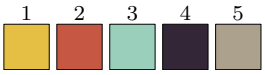
Dark24



FantasticFox1

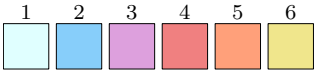


FantasticFox2



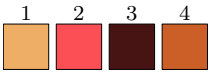
G10

GasFlame

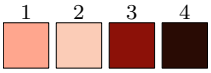


GeoRainbow

GrandBudapest1

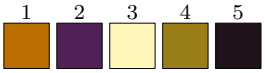


GrandBudapest2

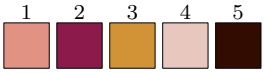


GrandBudapest3

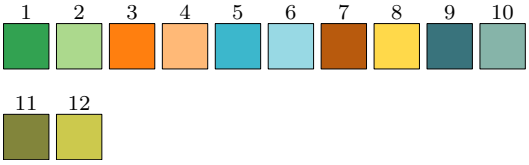
GrandBudapest4



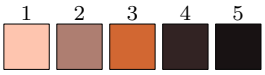
GrandBudapest5



GreenOrange



IsleOfDogs1



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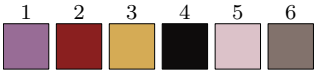

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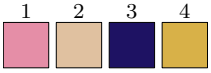
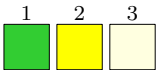
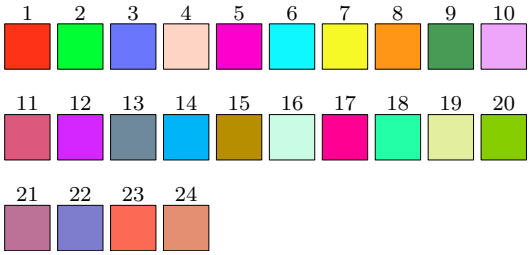


Lemon

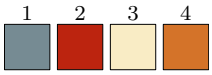


Light24



Margot1

Margot2



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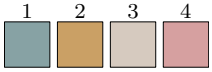
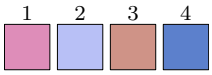
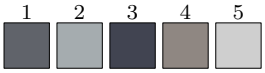


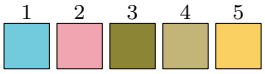
Mendl



MidGray

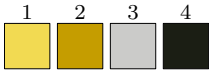


Moonrise1

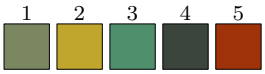


Moonrise2

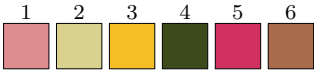
Moonrise3



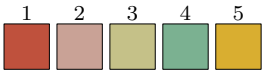
Moonrise4



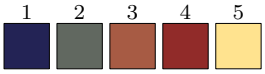
Moonrise5



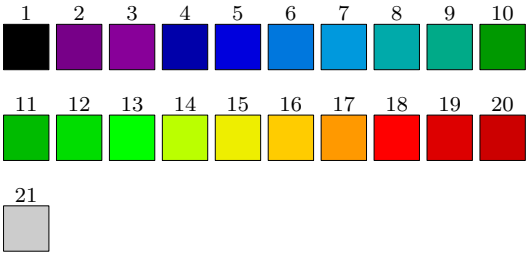
Moonrise6



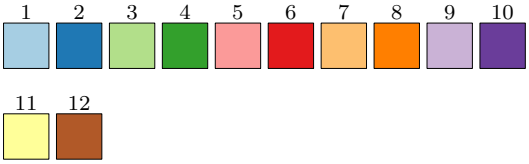
Moonrise7



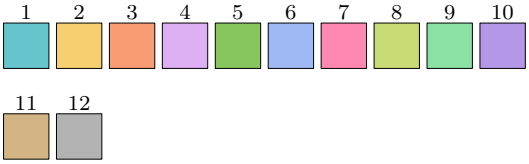
NipySpectral



Paired



Pastel



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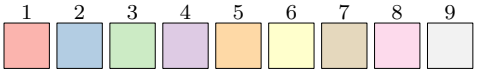

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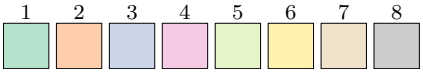
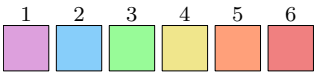
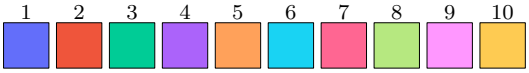


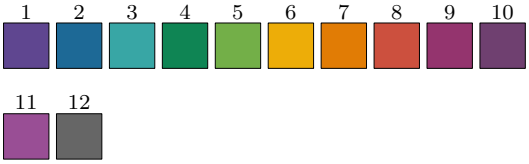
PastelRainbow



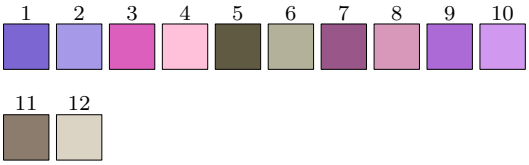
Plotly



PrismCC

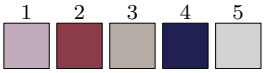


PurpleGray



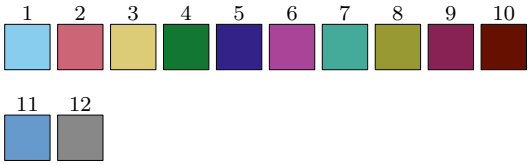
Royal1

Royal2

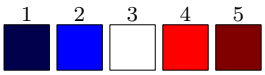


Royal3

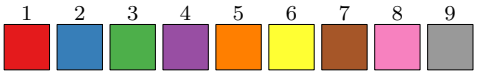
Safe



Seismic



Set1



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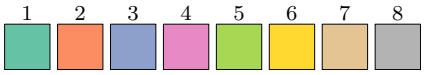

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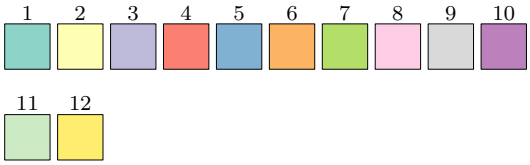
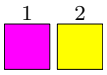
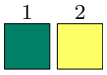


ShiftRainbow

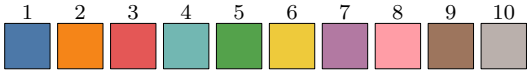
Spring



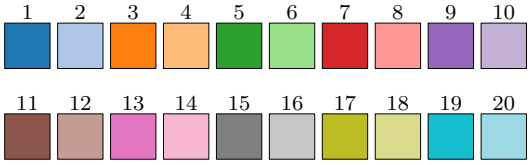
Summer



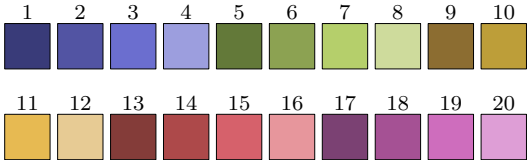
T10



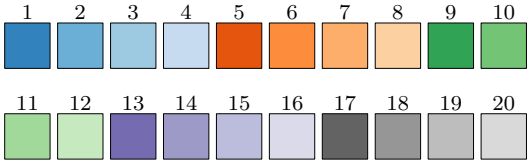
Tab20



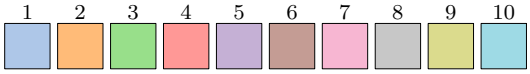
Tab20b



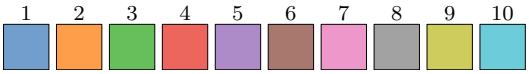
Tab20c



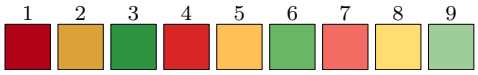
TableauLight



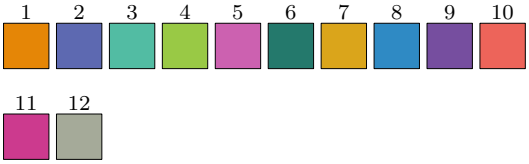
TableauMedium



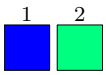
TrafficLight



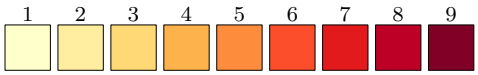
Vivid



Winter



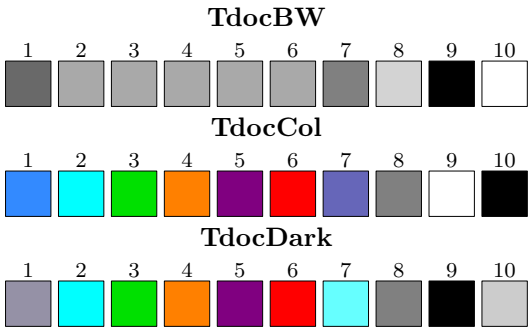
YlOrRd



Zissou

Semantic palettes

Semantic palettes assign colors to categories — such as UI states, system status, or pedagogical elements — to convey immediate meaning.



Sequential palettes

Sequential palettes represent ordered data using a single or multi-hue gradient with increasing lightness.

AfmHot

Algae

Amber

Amethyst

Amp

Apple

Arctic

Autumn

Binary

Blackbody

BlindFish

BlueRedPLY

Blues

Blues7

Blues10

BluesCP

BluGrn

BluYl

Bone

BRG

BrwnYl

Bubblegum

BuGn

BuPu

Burg

BurgYl

BurningGrass

BWR

Chroma

Cividis

Classic

CMRmap

Cool

CoolCP

Copper

Cosmic

Cube1

Cubehelix

Cubehelix1

Cubehelix2

Cubehelix3

CubeYF

DarkMint

Deep

Dense

Dusk

EarthCP

Eclipse

Electric

Ember

Emerald

Emrld

FallCM

Fire

Flamingo

Freeze

GasFlame

Gem

GeoRainbow

GhostLight

GistEarth

GistGray

GistHeat

GistNcar

GistRainbow

GistStern

GnBu

Gnuplot

Gnuplot2

Gothic

Gray

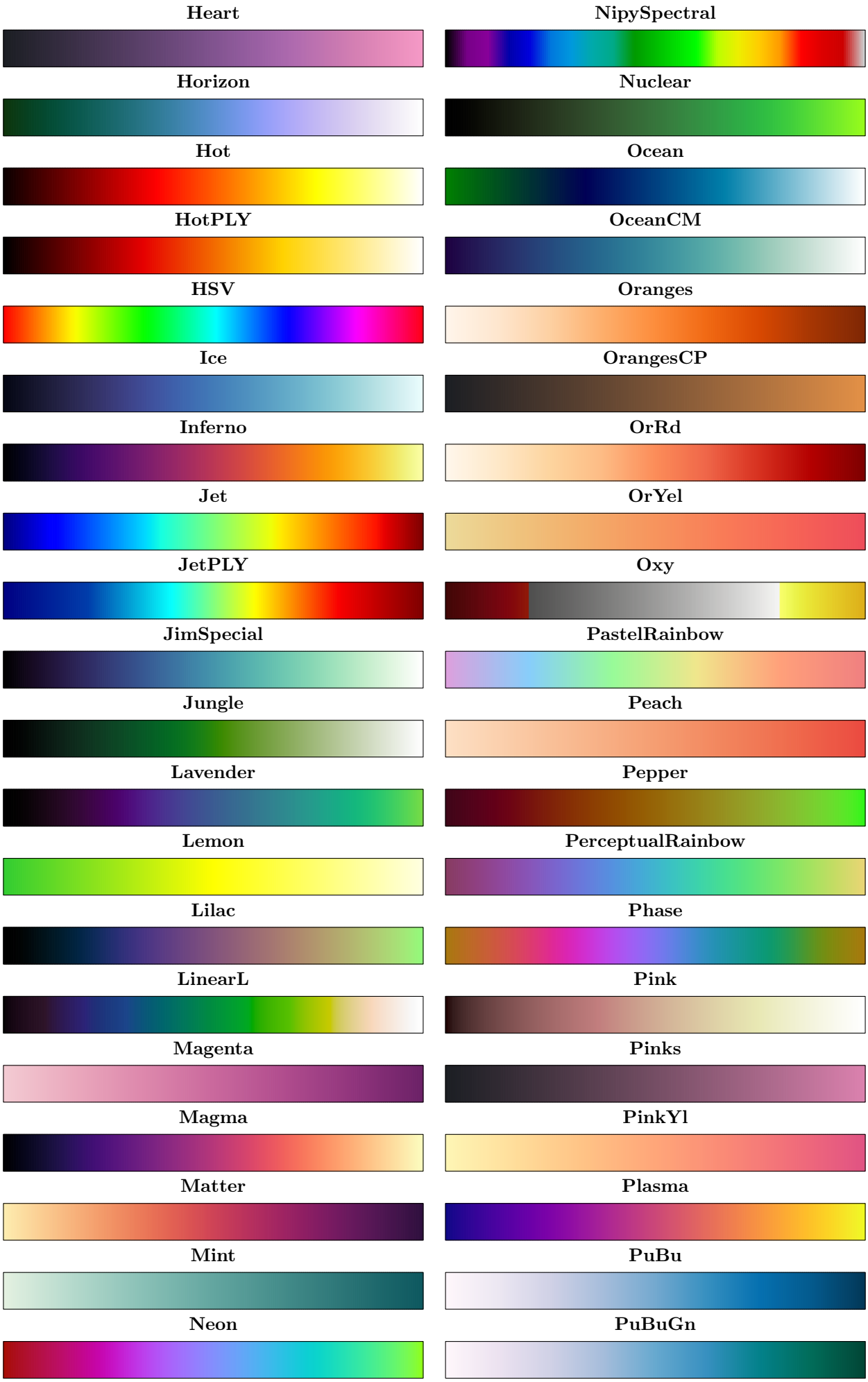
Greens

GreensCP

Greys

GreysCP

Haline


Heart
Horizon
Hot
HotPLY
HSV
Ice
Inferno
Jet
JetPLY
JimSpecial
Jungle
Lavender
Lemon
Lilac
LinearL
Magenta
Magma
Matter
Mint
Neon
NipySpectral
Nuclear
Ocean
OceanCM
Oranges
OrangesCP
OrRd
OrYel
Oxy
PastelRainbow
Peach
Pepper
PerceptualRainbow
Phase
Pink
Pinks
PinkYl
Plasma
PuBu
PuBuGn

PuRd
Purp
Purple
Purples
PurplesCP
PurpOr
Rainbow
RainbowPLY
Rainforest
RdPu
Red
RedOr
Reds
RedsCP
Sapphire
Seismic
Sepia
ShiftRainbow
Solar
Speed
Spring
Summer
SunBurst
Sunset
SunsetDark
Swamp
Teal
TealGrn
Teals
Tempo
Terrain
Thermal
Torch
Toxic
Tree
Tropical
Turbid
Turbo
Twilight
Viridis

Voltage

Warm

Water

Wind

Yellows

YlGn

YlGnBu

YlOrBr

YlOrRd